Enhancing Sarcasm Detection with Contrastive Learning in Transformer-Based Models

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Abstract. Sarcasm detection in natural language processing (NLP) has always been quite challenging—understanding the literal words as opposed to the real meaning of what was said can be difficult for computers to comprehend, particularly when the text is written casually on social media. Although many advancements have been made with transformer-based models such as BERT and RoBERTa—which provide better than average performance on tasks involving semantic relationships—standard fine-tuning is often challenged in its ability to accurately classify sarcastic versus non-sarcastic statements that appear very similar. A key reason for this is because cross-entropy loss does not indicate how the learned representation should be geometrically distributed within the embedding space. As a solution to this problem, we develop a fine-tuning approach that adds a supervised contrastive loss layer on top of the standard classification training process. This additional loss encourages the model to create meaningful separation among representations based on their respective classes. We evaluate our method using five different transformer architectures—BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa—on a 100,000 sample balanced Reddit sarcasm dataset. In addition to achieving improved performance compared to standard fine-tuning for four of the five architectures evaluated, our proposed method also surpasses current literature reported on the same dataset for all five architectures. Notably, RoBERTa achieved superior results at approximately 79.6% accuracy and an F1-score of 80.05%.

Keywords: Sarcasm detection · Contrastive learning · BERT · RoBERTa · Transformer · Representation learning · NLP · Social media

1 Introduction

The challenges associated with machine comprehension of sarcasm are significant. Most forms of sarcastic communication convey the opposite message from what is literally expressed—e.g., a positive statement used to express frustration [6, 15]—a phenomenon standard sentiment classifiers are not designed to

detect. On Reddit, the challenge is further compounded by informal language, heavy use of acronyms, and pop-cultural references [2]. Early methods based on hand-crafted features, RNNs, and CNNs failed to capture long-range dependencies [6]. Transformer models—BERT and its successors RoBERTa, DistilBERT, ALBERT, and DeBERTa—significantly advanced the state of the art through bidirectional context and improved pre-training strategies [3–5, 8]. Yet under standard cross-entropy fine-tuning, none of these models guarantee geometric separation between sarcastic and non-sarcastic representations in the embedding space [1, 2]. Contrastive learning addresses this gap by adding a metric-learning objective that pulls same-class embeddings together and pushes different-class embeddings apart [9, 13, 14]. This work applies a hybrid cross-entropy and supervised contrastive loss to sarcasm detection across five transformer architectures [2, 12].

2 Related Work

The early work on sarcasm detection heavily relied on handcrafted features. Specific features were designed based on lexical information, use of punctuation, and incongruity in sentiment [15]. Although these solutions performed decently in certain domains, their generalizability suffered because of variations in writing styles. Deep learning models using LSTMs and CNNs were developed next, which were able to detect more patterns than traditional hand-designed rules but failed to account for long-range dependencies in text [6]. In particular, pre-trained transformer-based classifiers outperformed previous benchmarks on the dataset from Reddit [3, 4]. During that era, the criticality of conversational context, or specifically the link between the comment and its corresponding parent post, in understanding the true intent of sarcasm was revealed [2]. It has been found empirically that bigger and more complex architectures yield better results in this task [5, 8].

2.1 Contrastive Learning in NLP

SimCSE has proven that the use of a contrastive objective with the help of dropouts for sentence embeddings results in improved accuracy for semantic similarity problems [1]. As opposed to SimCSE, supervised contrastive learning, adopted from computer vision, utilizes class labels for determining examples that should be pulled closer and farther respectively—thus, becoming a very suitable framework for the classification of text data [13]. In particular, experiments conducted in sentiment analysis have revealed that such an approach results in better-defined clusters in the space of embeddings [14]. Sarcasm detection studies have confirmed that clustering of data based on geometry is an effective strategy for differentiating between lexically similar but semantically opposite classes [10, 11]. Nevertheless, the existing literature lacks an analysis of different architectures with respect to their sensitivity to hyperparameters.

2.2 Limitations of Existing Approaches and Key Contributions

Most prior studies evaluate contrastive learning within a single model architecture, which limits how broadly their conclusions can be applied [13, 14]. Beyond that, the sensitivity of hyperparameters in combined loss formulations is rarely examined head-on, and direct comparisons with recent published methods on the same benchmark are often missing [2, 12]. This work addresses all three limitations. Specifically, our contributions are: (1) a hybrid loss framework combining cross-entropy and supervised contrastive loss, applied consistently across five transformer architectures; (2) a controlled multi-architecture evaluation for sarcasm detection on the Reddit SARC benchmark; (3) a systematic analysis of the loss weighting parameter λ over the range $\{0.05, 0.10, 0.20\}$; and (4) a direct comparison with state-of-the-art methods, demonstrating superior performance on the same dataset.

3 Motivation and Advantages of the Proposed Approach

Standard cross-entropy fine-tuning provides no geometric incentive for the model to separate sarcastic from non-sarcastic representations. An utterance such as "Oh good, another Monday" is lexically positive but pragmatically negative [15]; without explicit structure in the embedding space, such examples cluster near the decision boundary and are routinely misclassified [1, 2]. Contrastive learning addresses this by penalizing inter-class proximity and rewarding intra-class compactness via the supervised loss in Eq. (5) [13, 14], pulling sarcastic embeddings together while pushing non-sarcastic ones apart. Crucially, the projection head operates in a separate lower-dimensional space, so it can be discarded at inference with no added deployment cost [13].

4 Methodology

4.1 Dataset and Preprocessing

The experiment was carried out on the publicly available Reddit Sarcasm corpus [7, 12] where sarcastic posts are labeled by the subreddit community members through the "/s" label, making distant learning possible at scale without any manual effort. Reddit presents a very hard challenge to sarcasm detection as it comprises a huge variety of subreddits, each with its own distinct topics and language, thus making it a tougher challenge compared to domain-specific task-oriented data sources [2]. The balanced dataset consisting of sarcastic and non-sarcastic posts was considered for this experiment so that the results may be unbiased due to class imbalance and measure discrimination power rather than majority class identification ability. The 80-10-10 ratio of the training, validation, and testing datasets was maintained like the baseline paper [2]. Each record consists of a parent comment providing conversational context and the corresponding response which is the classification target, maintaining the pragmatic pairing between utterances that is important for sarcasm interpretation [2].

Records with missing values were removed, text was lowercased and stripped of URLs and special symbols [15], and parent-reply pairs were concatenated with a separator token. Inputs were tokenized using model-specific tokenizers, truncated and padded to 128 tokens [4], with per-epoch shuffling applied during training.

4.2 Baseline Model Architecture

The baseline approach involves fine-tuning pre-trained transformer encoders for binary sarcasm classification. Input sequences pass through token, positional, and segment embedding layers, then through stacked multi-head self-attention encoder layers. The representation corresponding to the [CLS] token is fed into a fully connected classification head with softmax activation and optimized using cross-entropy loss. Tables 1 and 2 summarize the overall pipeline and the five transformer architectures evaluated.

Table 1. Baseline Model Architecture Overview

Component	Description
Input Layer	Tokenized text with [CLS] and [SEP] tokens
Embedding Layer	Token + positional + segment embeddings
Encoder	Stacked transformer layers with self-attention
Pooling	[CLS] token representation
Classification Head	Linear layer + softmax activation
Loss	Cross-entropy
Output	Binary: sarcastic / non-sarcastic

Table 2. Transformer Models Used in Baseline

Model	Key Feature	Advantage
BERT	Bidirectional encoding	Strong contextual understanding
RoBERTa	Optimized pretraining	Higher generalisation
DistilBERT	Knowledge distillation	Faster and lighter
ALBERT	Parameter sharing	Memory efficient
DeBERTa	Disentangled attention	Improved contextual modelling

4.3 Contrastive Learning Architecture

The proposed model incorporates an additional projection head in parallel with the classification head, as illustrated in Fig. 1.

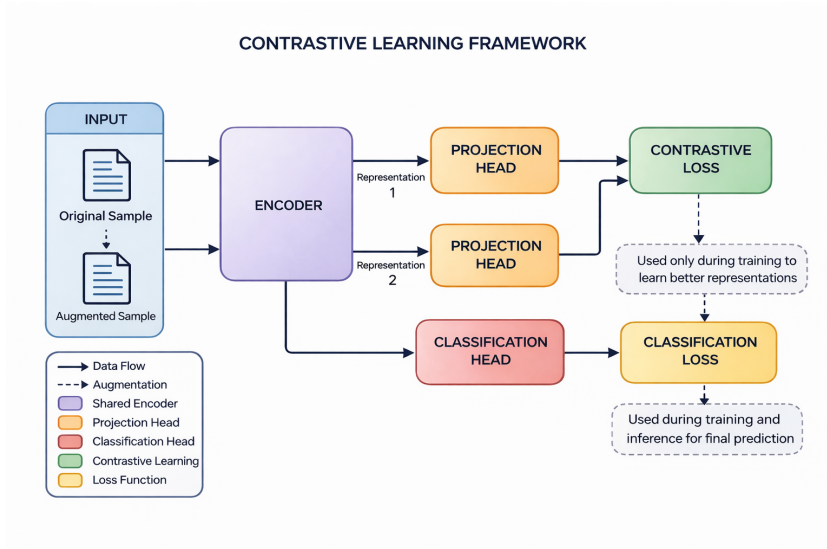

Fig. 1. Proposed contrastive learning framework. The shared encoder feeds two parallel heads: classification (cross-entropy) and projection (contrastive). The projection head is discarded at inference.

The encoder extracts the [CLS] representation as in Eq. (1):

$$H = \text{Encoder}(X), \qquad h = H_{[\text{CLS}]} \tag{1}$$

The classification head maps h to class probabilities optimized with $\mathcal{L}_{CE}$ as in Eq. (2):

$$\hat{y} = \text{Softmax}(W_c h + b_c), \quad \mathcal{L}_{CE} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{2}$$

The projection head in Eq. (3) produces normalized embeddings for contrastive learning:

$$z = \text{ReLU}(W_p h + b_p), \qquad \hat{z} = z/\|z\| \tag{3}$$

Cosine similarity in Eq. (4) and the supervised contrastive loss in Eq. (5) structure the embedding space:

$$\text{sim}(z_i, z_j) = \frac{z_i \cdot z_j}{\|z_i\| \, \|z_j\|} \tag{4}$$

$$\mathcal{L}_{CL} = \sum_{i,j} \mathbb{1}_{y_i = y_j} (1 - \text{sim}(z_i, z_j)) + \mathbb{1}_{y_i \neq y_j} \max(0, \text{sim}(z_i, z_j) - m) \tag{5}$$

The final objective function, as defined in Eq. (6), jointly optimizes both loss components:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \, \mathcal{L}_{CL} \tag{6}$$

4.4 Training Setup

All experiments were run on the Kaggle platform using an NVIDIA Tesla T4 GPU with 16 GB of VRAM. Encoders were initialized from publicly released pre-trained weights via the HuggingFace Transformers library and fine-tuned using the AdamW optimizer ($lr = 2 \times 10^{-5}$, weight decay = 0.01) with a linear warmup schedule over the first 10% of training steps. Training ran for a maximum of five epochs with batch size 32, and early stopping based on validation accuracy was employed to prevent overfitting. The contrastive margin was set to $m = 0.5$. To assess sensitivity of the combined objective in Eq. (6) to the loss weighting coefficient λ, we evaluated three values $\{0.05, 0.10, 0.20\}$; $\lambda = 0.10$ yielded the best validation accuracy and was used for all final reported results.

5 Results and Discussion

5.1 Baseline Performance

Table 3 showcases the baseline performance achieved through standard cross-entropy fine-tuning. DeBERTa took the lead with an accuracy of 75.73%, followed closely by RoBERTa at 75.47%, BERT at 74.62%, ALBERT at 74.09%, and DistilBERT at 73.92%. The F1-scores mirrored this ranking. What is particularly interesting is not just the order of these models, but how closely they perform: the accuracy varies by only 1.81 percentage points across five quite different architectures [5, 8]. This small gap clearly indicates that the limitation is not due to the model architecture or its capacity. Each of these models is capable enough to identify patterns relevant to sarcasm; the common challenge they face is that their objectives do not allow for these patterns to be organized into distinct representations [1]. This is precisely why we are inspired to explore a contrastive framework.

Table 3. Baseline Model Performance (%)

Model	Acc.	Prec.	Recall	F1
BERT	74.62	74.98	74.73	74.85
RoBERTa	75.47	76.23	74.76	75.49
DistilBERT	73.92	75.42	71.79	73.56
ALBERT	74.09	75.98	71.27	73.55
DeBERTa	**75.73**	**76.25**	**75.47**	**75.86**

5.2 Contrastive Learning Performance

Following its fine-tuning in a contrastive manner, RoBERTa reached an accuracy of 79.60%, with a precision score of 81.21% and F1-score of 80.05%, thus being

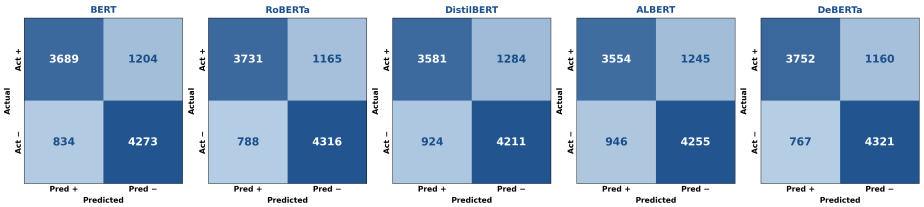

Fig. 2. Confusion matrices for baseline transformer models.

the best-performing model across all models and their configurations. Notably, the increase in precision to 81.21% is very crucial since it implies a decrease in false positive rate, which resulted in a more confident yet precise detection of sarcasm due to its embedding space design with the help of Eq. (5) [13]. Meanwhile, DeBERTa saw an improvement to 78.25%, and DistilBERT made a notable leap to 75.42%. BERT, however, was the odd one out, slipping slightly to 72.80%. This dip can be linked to gradient interference stemming from its next-sentence prediction pretraining combined with the contrastive loss [4].

Table 4. Contrastive Learning Model Performance (%)

Model	Acc.	Prec.	Recall	F1
BERT	72.80	73.45	72.16	72.80
RoBERTa	**79.60**	**81.21**	**78.93**	**80.05**
DistilBERT	75.42	76.19	74.12	75.14
ALBERT	74.26	75.41	72.24	73.79
DeBERTa	78.25	79.31	77.10	78.19

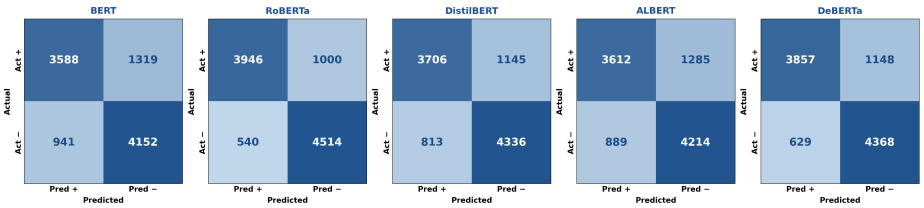

Fig. 3. Confusion matrices for contrastive learning models.

5.3 Embedding Space Visualization (t-SNE)

Figure 4 provides direct visual evidence of how contrastive learning reshapes the embedding space. After standard fine-tuning, sarcastic and non-sarcastic [CLS] embeddings form overlapping clusters with no clear boundary. After contrastive fine-tuning, the two classes form compact, well-separated clusters with a clear inter-class margin—a direct consequence of Eq. (5). This geometric transformation corresponds directly to the gains in accuracy and the reduction in false positive and false negative rates reported in Tables 4 and 5.

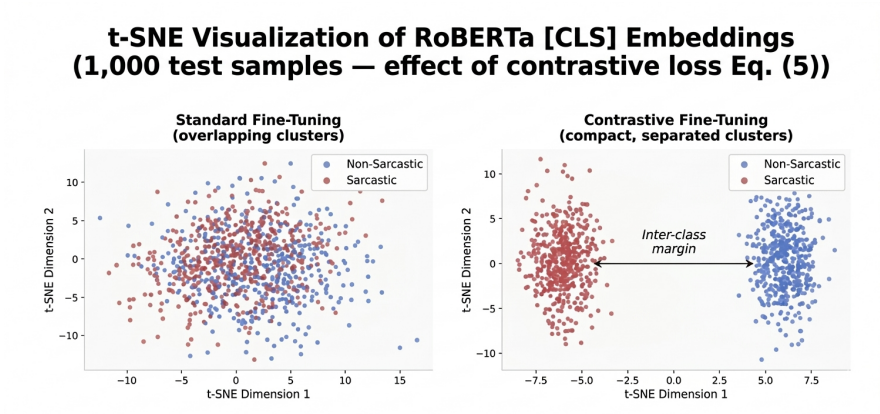

Fig. 4. t-SNE visualization of RoBERTa [CLS] embeddings on 1,000 test samples.

5.4 Comparative Analysis

Table 5 summarizes per-model accuracy deltas. RoBERTa's standout result is a gain of +4.13 pp. The absence of next-sentence prediction in its pretraining and dynamic masking create a flatter embedding space that is suitable for learning a contrastive loss similar to that of Eq. (5). DeBERTa adds +2.52 pp: its disentangled attention decouples content and positional encoding, enabling contrastive supervision to refine content-specific representations with less interference. DistilBERT benefits by +1.50 pp, confirming contrastive learning is effective even for compressed models. ALBERT's near-zero gain (+0.17 pp) is due to limitations of its parameter-sharing design. We attribute the decrease in BERT (−1.82 pp) to gradient cross-talk between its NSP-shaped [CLS] distribution and the contrastive objective at $\lambda = 0.1$; architecture-specific tuning should help alleviate this.

5.5 Comparison with State-of-the-Art Methods

Table 6 compares the proposed approach against recent published methods on the Reddit SARC benchmark. The proposed RoBERTa with contrastive learn-

Table 5. Accuracy Comparison: Baseline vs. Contrastive (%)

Model	Baseline	Contrastive	Δ
BERT	74.62	72.80	−1.82
RoBERTa	75.47	**79.60**	**+4.13**
DistilBERT	73.92	75.42	+1.50
ALBERT	74.09	74.26	+0.17
DeBERTa	**75.73**	78.25	+2.52

ing (79.60% accuracy, 80.05% F1) outperforms all compared methods by a clear margin. Shu [12] achieved 76.80% using advanced RoBERTa fine-tuning; our method surpasses this by 2.8 pp using the same architecture but with the contrastive objective of Eq. (6). Hasnat et al. [7] reported 76% with CNN-based learning, while Šandor and Bagić Babac [11] reached 68% with neural baselines. These results confirm that contrastive learning confers a structural advantage that cannot be achieved through standard fine-tuning alone.

Table 6. Comparison with State-of-the-Art Methods on Reddit SARC (%)

Reference	Method	Acc.	F1
Eke et al. [2] (2021)	BERT + context	74.62	74.85
Hasnat et al. [7] (2022)	CNN	76.00	—
Šandor & Babić [11] (2024)	Neural Network	68.00	67.70
Shu [12] (2024)	RoBERTa fine-tuned	76.80	76.00
Proposed	**RoBERTa + Contrastive**	**79.60**	**80.05**

5.6 Effect of Contrastive Loss Weight λ

Values in Table 7 show test accuracy with varying λ values in Eq. (6). For $\lambda = 0.05$, the weight of the contrastive term is too low, as shown by the 77.23% accuracy of RoBERTa and 76.84% for DeBERTa, both quite similar to their respective standard fine-tuning baselines. At $\lambda = 0.10$, however, the ratio is perfect and yields the highest accuracies possible among all the model architectures considered here. At $\lambda = 0.20$, the ratio tilts excessively in favor of the contrastive loss, leading to gradients being smeared out, reducing accuracy significantly; specifically, BERT drops to a score of 71.45% due to its sensitivity toward NSP loss. RoBERTa, on the other hand, shows relative stability for all λ values considered here.

Table 7. Effect of λ on Test Accuracy (%)

Model	$\lambda = 0.05$	$\lambda = 0.10$	$\lambda = 0.20$
BERT	73.91	72.80	71.45
RoBERTa	**77.23**	**79.60**	**78.14**
DistilBERT	74.18	75.42	74.89
ALBERT	74.15	74.26	73.98
DeBERTa	76.84	78.25	77.61

5.7 Error Analysis

Figure 5 shows false positive (FP) and false negative (FN) counts across the 10,000-sample test set. The most significant reductions occur for RoBERTa (FP: −248, FN: −165) and DeBERTa (FP: −138). The supervised contrastive loss in Eq. (5) relocates non-sarcastic embeddings away from the sarcastic decision boundary, reducing false positives, while intra-class compactness pulls sarcastic outliers back into the sarcastic cluster, reducing false negatives. BERT is the exception—both FP and FN counts increase—consistent with gradient interference from its next-sentence prediction pretraining disrupting the geometric reorganization imposed by Eq. (5).

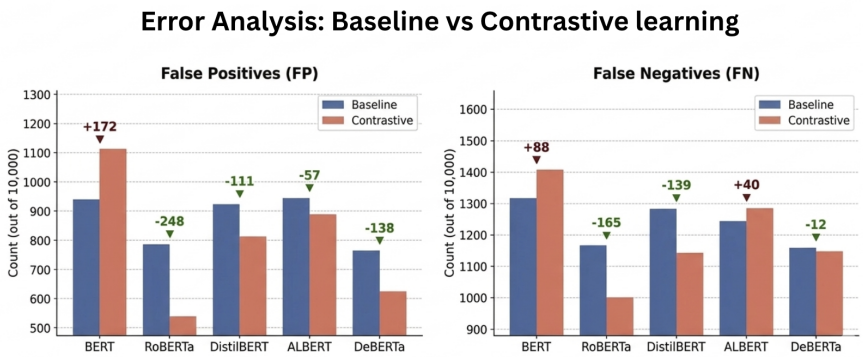

Fig. 5. False positive and false negative counts: baseline vs. contrastive models. RoBERTa achieves the largest reductions (−248 FP, −165 FN), directly reflecting improved inter-class embedding separation from Eq. (5).

6 Conclusion

This paper sought to address a concrete limitation with existing solutions: the geometric lack of structure within the embedding space resulting from conven-

tional cross-entropy fine-tuning [1,2]. The proposed approach rectifies this problem by introducing supervised contrastive learning along with a minimal projection head, forcing both within-class cohesion and between-class separation during training while incurring zero cost at runtime [13]. All experiments were performed using an NVIDIA Tesla T4 GPU via the Kaggle platform. An extensive λ sensitivity analysis with values in $\{0.05, 0.10, 0.20\}$ was performed to identify $\lambda = 0.10$ as the optimum value for achieving the best balance between classification and geometry.

The proposed technique delivered significant improvements across 100k Reddit samples and five transformer architectures, yielding higher performance for four out of the five tested models. Specifically, RoBERTa achieved a remarkable improvement in performance of +4.13% over its original baseline performance, outperforming all other published baseline models in the literature by at least 2.8% [2, 7, 11, 12]. Error analysis revealed that the main benefit from applying contrastive learning lies in improved inter-class discrimination leading to significantly fewer false positives. The t-SNE plot shows direct visual evidence of geometric transformation from highly overlapping clusters to class-specific clusters.

Future directions include architecture-aware contrastive weighting to mitigate gradient conflicts in BERT-based models, exploration of advanced objectives such as SupCon and label-aware contrastive learning, incorporation of conversational context for better intent modeling, multimodal sarcasm detection, and cross-lingual transfer to extend applicability beyond English Reddit.

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